FUN-3.A.1 Direct application of the definition of the derivative and specific rules can be used to calculate the derivative for functions of the form f(x) = x^n

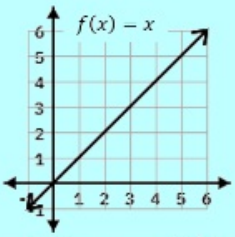
FUN-3.A.2 Sums, differences, and constant multiples of functions can be differentiated using derivative rules.

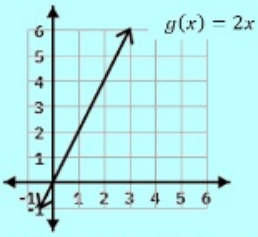
FUN-3.A.3 The power rule combined with sum, difference, and constant multiple properties can be used to find the derivatives for polynomial functions.

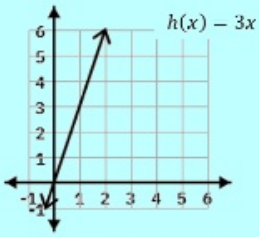
2-5 & 2-6 Power Rule and Basic Derivative Rules

Get Ready...

Find the following:







$f'(3) = 1 \quad f'(5) = 1$

$f'(x) = 1$

$g'(3) = 2 \quad g'(5) = 2$

$g'(x) = 2$

$h'(3) = 3 \quad h'(5) = 3$

$h'(x) = 3$

Reminder: Given a function $f(x)$, the derivative of $f(x)$ is denoted by $f'(x)$. We can find $f'(x)$ using the definition of the derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

As you probably remember, this process can be tedious. We certainly wouldn't want to use this process to find all derivatives throughout the course! Thankfully, we will learn several derivative rules that have been derived from the limit process.

Derivative of a constant	$\frac{d}{dx}(c) =$ 0	**Derivative of a constant times x**	$\frac{d}{dx}(cx) =$ c

Power Rule	If $f(x) = x^n$, then $f'(x) =$ nx^{n-1} Important notes: **#1:** This ONLY works for polynomials (powers of x). **#2:** Make sure that x is in the **numerator** before using the Power Rule!!!

$f(x) = x^5 \quad f'(x) = 5x^4$

$g(x) = 3x^4 \quad g'(x) = 12x^3$

$h(x) = \frac{1}{2}x^6 \quad h'(x) = 3x^5$

The Constant Multiple Rule	If c is a constant and f is a differentiable function, then $$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$$
The Sum Rule	If f and g are both differentiable functions, then $$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$
The Difference Rule	If f and g are both differentiable functions, then $$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$$

The sum and difference rules can be extended to more terms.

Example 1: Given $f(x)$ below, find $f'(x)$.

a. $f(x) = x^3$

$f'(x) = 3x^2$

b. $f(x) = -3x^2$

$f'(x) = -6x$

c. $f(x) = 5x$

$f'(x) = 5$

d. $f(x) = 5x^{-2}$

$f'(x) = -10x^{-3}$

$= \frac{-10}{x^3}$

e. $f(x) = 7$

$f'(x) = 0$

f. $f(x) = \frac{2}{3}x^4$ $4 \cdot \frac{2}{3}$

$f'(x) = \frac{8}{3}x^3$

$= \frac{8x^3}{3}$

Example 1: Given $f(x)$ below, find $f'(x)$.

g. $f(x) = \frac{x^4}{6} = \frac{1}{6}x^4$

$f'(x) = \frac{2}{3}x^3$

$= \frac{2x^3}{3}$

h. $f(x) = \frac{3}{2x^4} = \frac{3x^{-4}}{2}$

$= \frac{3}{2}x^{-4}$

$f'(x) = -6x^{-5}$

$= \frac{-6}{x^5}$

i. $f(x) = \frac{3x}{5} \leftarrow \frac{3}{5}x$

$f'(x) = \frac{3}{5}$

j. $f(x) = \frac{1}{x} + 2x^2$

$= x^{-1} + 2x^2$

$f'(x) = -1x^{-2} + 4x$

$= \frac{-1}{x^2} + 4x$

k. $f(x) = \frac{1}{x^2} - \frac{2}{x^3} + 3x - 7$

$= x^{-2} - 2x^{-3} + 3x - 7$

$f'(x) = -2x^{-3} + 6x^{-4} + 3$

$= \frac{-2}{x^3} + \frac{6}{x^4} + 3$

l. $f(x) = \pi^3$

$f'(x) = 0$

Working with Radicals	In calculus, we will always turn **<u>radical expressions</u>** into **<u>power</u>** form. **Remember**: $f(x) = \sqrt{x}$ can be written as $f(x) = x^{\frac{1}{2}}$ **Additionally**: $f(x) = \sqrt[n]{x}$ can be written as $f(x) = x^{\frac{1}{n}}$

$\sqrt[5]{x} = x^{1/5}$

$\sqrt[3]{x^2} = x^{2/3}$

Example 2: Given $g(x)$ below, find $g'(x)$.

a. $g(x) = \sqrt{x}$

$= x^{1/2}$

$g'(x) = \frac{1}{2}x^{-1/2}$

$= \frac{x^{-1/2}}{2}$

$= \frac{1}{2x^{1/2}} = \frac{1}{2\sqrt{x}}$

b. $g(x) = 2\sqrt[5]{x}$ $= 2x^{1/5}$

$g'(x) = \frac{2}{5}x^{-4/5}$

$= \frac{2x^{-4/5}}{5}$

$= \frac{2}{5x^{4/5}}$

c. $g(x) = \frac{4}{\sqrt{x}}$ $= 4x^{-1/2}$

$g'(x) = -2x^{-3/2}$

OR $\frac{-2}{x^{3/2}}$

d. $g(x) = \sqrt{x^3}$

$= x^{3/2}$

$g'(x) = \frac{3x^{1/2}}{2}$

e. $g(x) = \frac{\sqrt{x}}{5}$

$= \frac{x^{1/2}}{5} = \frac{1}{5}x^{1/2}$

$g'(x) = \frac{1}{10}x^{-1/2}$

$= \frac{x^{-1/2}}{10}$

$= \frac{1}{10x^{1/2}} = \frac{1}{10\sqrt{x}}$

f. $g(x) = \sqrt[3]{x} - \frac{1}{\sqrt[4]{x}}$

$= x^{1/3} - x^{-1/4}$

$g'(x) = \frac{1}{3}x^{-2/3} + \frac{1}{4}x^{-5/4}$

$= \frac{x^{-2/3}}{3} + \frac{x^{-5/4}}{4}$

Example 2: Given $g(x)$ below, find $g'(x)$.

g. $g(x) = \sqrt[3]{e}$

$g'(x) = 0$

h. $g(x) = \frac{2}{\sqrt[5]{x^3}} = 2x^{-3/5}$

$g'(x) = \frac{-6}{5} x^{-8/5}$

$= \frac{-6x^{-8/5}}{5}$

i. $g(x) = -2\sqrt{x} + 3$

$= -2x^{1/2} + 3$

$g'(x) = -1x^{-1/2}$ ☺

$= \frac{-1}{x^{1/2}} = \frac{-1}{\sqrt{x}}$

j. $g(x) = \frac{1}{x} - \sqrt{x} + 5x$

$= \frac{1}{x} - x^{1/2} + 5x$

$g'(x) = \frac{-1}{x^2} - \frac{1}{2}x^{-1/2} + 5$

$= \frac{-1}{x^2} - \frac{x^{-1/2}}{2} + 5$

Horizontal Tangent Lines	The slope of a horizontal tangent line is zero. To find the x-value where a function has a horizontal tangent line, we set the derivative equal to zero and solve for x: $f'(x) = 0$ ***Caution:** If you are asked for the <u>**point**</u> where the tangent line is horizontal, you need to find an ordered pair! (Use the original equation $f(x)$.

Example 3: Find the point(s) on the curve $y = x^4 - 2x^2 + 4$ where the tangent line is horizontal.

$y' = 4x^3 - 4x$

$4x^3 - 4x = 0$

$4x(x^2 - 1) = 0$

$4x = 0 \quad x^2 - 1 = 0$

$x = 0 \quad x^2 = 1$

$x = \pm 1$

$y(0) = 0 - 0 + 4 = 4$

$y(1) = 1 - 2 + 4 = 3$

$y(-1) = 1 - 2 + 4 = 3$

$y(0) = 4$

$y(1) = 3$

$y(-1) = 3$

Normal Lines	A **Normal Line** to a curve at a point is the line **perpendicular** to the tangent line at that point.

Slopes of ⊥ lines

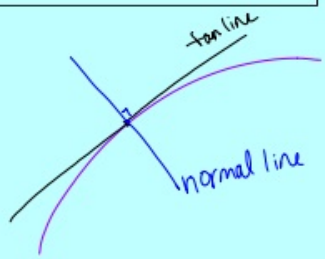
negative reciprocal

neg. rec.

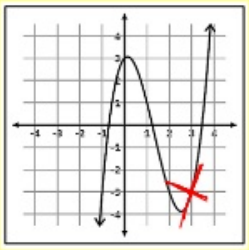
$\frac{1}{4} \rightarrow -4$

$-8 \rightarrow \frac{1}{8}$

$-\frac{2}{5} \rightarrow \frac{5}{2}$



Example 4: Find the equation of the normal line to the graph of $f(x) = x^3 - 4x^2 + x + 3$ at $x = 3$.



$f(3) = 27 - 36 + 3 + 3 = -3$

$f'(x) = 3x^2 - 8x + 1$

$m_{tan} = f'(3) = 27 - 24 + 1 = 4$

$m_{norm} = -\frac{1}{4}$

$EQ_{norm}: y + 3 = -\frac{1}{4}(x - 3)$

Differentiability with piecewise functions

Example 5: Is the function $f(x) = \begin{cases} 5x^2 + 3x + 2 & x < -1 \\ -7x - 3 & x \geq -1 \end{cases}$ differentiable at $x = -1$?

Continuous at $x=-1$?

$f(-1) = 7 - 3 = 4$ $\quad \lim_{x \to -1} f(x) = 4$

$\lim_{x \to -1^-} f(x) = \lim_{x \to -1^-} (5x^2 + 3x + 2) = 5 - 3 + 2 = 4$

$\lim_{x \to -1^+} f(x) = \lim_{x \to -1^+} (-7x - 3) = 4$

So $f(x)$ is cont. at $x = -1$...

$$f'(x) = \begin{cases} 10x + 3 & x < -1 \\ -7 & x \geq -1 \end{cases}$$

$f'(-1)$ from the left: $-10 + 3 = -7$
$f'(-1)$ from the right: -7
Same

So yes, $f(x)$ is differentiable at $x = -1$

Example 6: What values of a and b would make the function $f(x) = \begin{cases} x^2 - ax + 2 & x < 3 \\ x + b & x \geq 3 \end{cases}$ differentiable at $x = 3$?

differentiable $\longrightarrow$ continuous at $x=3$

$f(3) = 3 + b$

$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^-} (x^2 - ax + 2) = 9 - 3a + 2 = 11 - 3a$

$\lim_{x \to 3^+} f(x) = \lim_{x \to 3^+} (x + b) = 3 + b$

$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x)$

$11 - 3a = 3 + b$

$11 - 15 = 3 + b$

$-4 = 3 + b$

$-7 = b$

f is differentiable...

$f'(x) = \begin{cases} 2x - a & x < 3 \\ 1 & x \geq 3 \end{cases}$

$f'(3)$ from left $= f'(3)$ from right

$6 - a = 1$

$-a = -5$

$a = 5$